

Successful Calls, Insufficient Evidence: Scope and Coverage in Tool-Using Agents

Research manuscript; not submission-ready. The empirical studies reported here are complete. The core saved-record artifact passed Windows and Linux reanalysis. The later annotation-sensitivity extension has a separate local verification record and is not included in that historical artifact. Independent answer adjudication and complete public artifact release remain unfinished. The preceding manuscript is preserved verbatim in `study_record.md` as a detailed study record.

Abstract

Successful tool execution does not establish that an agent has enough evidence to answer a question. We study public relational-tool tasks, separating execution validity, answer correctness and identification by the observed trace. A fixed coverage reminder has mixed domain effects. In a matched 240-episode executor control, accepting sequential call batches gives Qwen2.5 twelve additional final answers but only one net additional correct answer under fixed SQL interpretations. A separate SmolLM3 study produces six correct answers among 220 scored executions. A larger Qwen3 checkpoint scores 35/55 versus 38/55 with the reminder on the known tasks. A prospective 420-episode expansion across four further databases yields original/reminder counts of 24/49 versus 20/49, 14/49 versus 15/49, and 27/49 versus 29/49 across three checkpoints. All conditional paired intervals include zero; omitting Disney reverses both positive aggregate differences. Forty-three input-budget failures and incomplete requested lists further delimit interpretation. On two prospectively length-selected tasks, larger generation budgets complete four previously truncated answers but leave eight incomplete responses unchanged. Full-transcript database replays provide two examples where a correct original answer is not identified under an explicit admissible intervention. One retrieves every requested output value yet applies an incorrect predicate. A bounded price-swap search adds no correct-answer witness. These results support reporting execution, correctness and evidence sufficiency separately, with explicit limits on counterfactual admissibility and resource budgets. They are diagnostic results on public training tasks, reviewed by one unblinded assistant; they establish neither official benchmark scores, population failure prevalence nor a new controller's superiority.

1. Introduction

A tool response is both a computational result and evidence for an agent's next decision. These roles can diverge. A count operation may execute correctly on an incorrectly scoped table. A preview may contain only true values while omitting part of a requested list. Even retrieving every output value can leave the answer dependent on an unchecked predicate. Evaluating call validity misses the first two problems; evaluating answer agreement can miss the third.

Relational tools make these distinctions inspectable. A server keeps complete relations behind handles while exposing schemas, row counts and previews. Filtering, sorting, aggregation or retrieval can recover further evidence, but large observations consume context and complete answers can exceed output limits. Evaluation must distinguish what an interface could reveal, what an agent actually observes and what its output channel can express.

Accidental correctness and trace-conditioned determinacy are established concerns in [semantic SQL evaluation](#) and [database access control](#). Our contribution is an empirical diagnosis in interactive tool use: matched controls expose different failure sources, and complete observation replays provide checkable, explicitly local counterexamples. We do not introduce determinacy, pagination or a universal grounding verifier.

We ask three questions. **RQ1:** Does a fixed scope-and-coverage reminder transfer beyond its development database? **RQ2:** How much do an executor restriction and the generation budget account for apparent answer failures? **RQ3:** Can a correct answer remain unidentified by the complete observed trace under a declared class of alternative database states?

The prospective comparison answers RQ1 with heterogeneous effects and sign reversals under domain removal. Executor and output-budget controls answer RQ2 differently: more final responses yield little net correctness gain,

whereas four selected lists exhibit exact token-prefix evidence of output truncation. Two native-tool replays answer RQ3 by preserving all target observations despite changing the required answer. Retained inadmissible and unsuccessful searches limit generalization from those positive examples.

2. Evidence identification

Let D denote a database, q a fixed interpretation of the requested computation, and h the agent's observable history. For a declared admissibility condition A , let $C_A(h)$ contain states that satisfy A and reproduce h under the actual tool semantics. The answer is identified by h within A precisely when q has the same value on every state in $C_A(h)$. This concerns information in the trace, not the absence of useful model priors or the correctness of its final answer.

A replay witness is a D' in A with $q(D')$ different from $q(D)$ but exactly the same observations under the recorded action sequence. We compare initialization, ordered tool definitions, complete tool content and error flags. Equality of selected fields or preview lengths is insufficient. For a deterministic policy with identical fresh model state, observation equality preserves the next action inductively. A stochastic policy additionally requires coupled random choices; independently sampled trajectories need not match.

Admissibility carries substantive assumptions. A country-label edit can preserve SQLite constraints while violating geographical knowledge used to interpret a question. Our retained examples therefore declare which status field may vary and which cells stay fixed. A witness refutes identification within that class; failure to find one in a bounded search certifies nothing. This is an application of established determinacy ideas, not a new theorem.

We separately record interaction termination, requested-value agreement and local replay witnesses. A final response is not necessarily a correct answer, and a correct value need not have a correct explanation. A preview can suffice for positive membership while failing an exhaustive list; an exact aggregate can suffice without reading every row, provided its scope is correct. No aggregate grounding rate is inferred from the selected replay examples.

3. Experimental design

3.1 Environment, checkpoints and controls

We use public capability-1 training tasks from VAKRA, server commit `c9e82dfe46aee016d2bfe3a0c8fed652760e4c0c` and dataset revision `1388b9f1aaed887a73957eba651a6c2034010476`. Unchanged native MCP tools operate on local database copies. Initialization uses the upstream reference-derived relation or join, an oracle setup convention rather than agent planning. Agents receive questions, tool schemas and the public initial preview, but no reference program, SQL card or answer. VAKRA's own evaluation already considers trajectories and grounding; our diagnostic labels are not its [official scores](#).

Every target starts with fresh system and user messages. Server registration can persist within a worker, so replays preserve warmup and preceding tool order. A logged universe switch activates dynamic getters before evaluation, and tool schemas are refreshed. This workaround changes no data semantics.

The original condition retains the upstream system-prompt method. The reminder appends instructions to distinguish previews from full columns, check requested predicates, use filtered aggregates where appropriate, and disclose incomplete evidence. Its exact text is in `autolab/vakra_native.py` and Appendix A below. It was developed on world examples, then fixed for subsequent comparisons; it is a simple instruction control, not a learned or verified controller.

The expansion uses Qwen3-4B-Instruct-2507, Qwen2.5-7B-Instruct and Qwen3-30B-A3B-Instruct-2507 with local Transformers inference and greedy decoding. Checkpoint files, dependencies, source and input identities are hash-pinned. Smaller checkpoints use one A40 per worker; Qwen3-30B-A3B uses two with all layers resident on GPUs. This is a checkpoint comparison, not a controlled scaling experiment.

Primary runs allow twenty model attempts, twenty tool attempts, 512 new tokens per generation and 32,768 input tokens. Over-limit inputs terminate without silent truncation. The sequential executor accepts valid call batches

and counts every executed attempt; unknown tools or oversized batches are rejected before partial execution. The earlier strict executor rejects multi-call generations, following the prompt's one-call instruction. Tool errors, malformed outputs and budget failures are retained.

3.2 Selection and study sequence

Table 1. Study sequence and number of model episodes. Overlapping studies are not independent task samples. CPU tool replays and evaluator-authored acquisition controls are not new model episodes.

Study	Task set	New model episodes	Role
Initial observations	4 world queries, 2 checkpoints	8	Development; one OOM
Matched development	12 world queries, 2 checkpoints, 2 prompts	48	Includes the initial queries
Strict replication	60 queries, 3 other databases, 2 checkpoints, 2 prompts	240	Fixed prompt transfer
Sequential executor	Same 60 queries and conditions	240	Post-outcome protocol control
SmolLM3-3B	Same 60 queries, 2 prompts, 2 executors	240	Checkpoint sensitivity
Qwen3-30B-A3B	Same 60 queries, 2 prompts	120	Known-task extension
Prospective expansion	70 queries, 4 further databases, 3 checkpoints, 2 prompts	420	Primary transfer evidence
Larger generation budget	2 expansion tasks, 3 checkpoints, 2 prompts	12	Registered length sensitivity

The initial replication selects the three smallest previously untested non-world capability-1 databases by file size: `computer_student`, `cars` and `book_publishing_company`, with twenty released inputs each. The expansion excludes those and `world`, retains eligible files at most 5,000,000 bytes, and selects four domains in seeded hash order. A fixed task hash ordering selects at most twenty examples per domain: `cookbook` (20), `Disney` (20), `genes` (10) and `ice_hockey_draft` (20). No task is replaced after answer review. These are new to our experiments, not necessarily to pretraining. SmolLM3's earlier floor motivated its omission from the expansion, so checkpoint selection was informed by development.

3.3 Scoring and uncertainty

Before expansion predictions, reference SQL and task wording were inspected to fix interpretation cards and an ambiguity mask. Twenty-one of seventy tasks have recorded wording/reference conflicts or unresolved ambiguity. All seventy remain in execution and cost accounting; forty-nine enter each answer comparison, giving 294 scored and 126 separately reported ambiguous executions. Long answers and failed runs receive no new exclusion. Missing answers count as unsuccessful in the fixed denominator. Earlier three-domain comparisons analogously retain 55 of 60 tasks in their frozen SQL subtotal.

One unblinded assistant read every complete response or recorded absence and assigned requested-value labels against the cards. Correct values can be credited despite faulty explanations; annotation reasons and sensitivity labels disclose those cases. There is no independent human adjudication. Identical final text in the executor comparison reuses its answer label; changed responses are reviewed. Label reuse does not establish grounding.

We pair prompt outcomes by task and preserve domain-specific scored counts when resampling. Expansion intervals use the exact finite distribution of the ordinary within-domain paired bootstrap; earlier intervals use 10,000 resamples. These post-completion descriptive calculations condition on the observed domains and assistant labels. They cover neither annotation error, unseen domains nor generation seeds. Repeated runs and checkpoints

are not independent tasks. Detailed protocols and the full study sequence remain in `study_record.md`.

4. RQ1: Mixed transfer of the coverage reminder

4.1 Prospective task and domain comparisons

Table 2. Requested-value matches against frozen interpretations. The genes floor concerns only three scored tasks and does not establish equivalence.

Domain	Scored tasks per arm	Qwen3-4B original / reminder	Qwen2.5-7B original / reminder	Qwen3-30B-A3B original / reminder
Cookbook	16	9 / 8	7 / 7	11 / 11
Disney	18	12 / 9	5 / 7	12 / 15
Genes	3	0 / 0	0 / 0	0 / 0
Ice hockey	12	3 / 3	2 / 1	4 / 3
Combined	49	24 / 20	14 / 15	27 / 29

All 24 workers exit successfully, producing the exact 420 registered records and 210 verified initial prompt pairs. The complete archive retains 364 final responses, forty protocol errors and nine finals at the generation ceiling. The fixed 21-task ambiguity mask excludes 126 executions from answer accuracy while retaining all costs. Thus each arm has 49 scored tasks across the same four databases.

Table 3. Paired reminder effects on the 49 scored expansion tasks.

Checkpoint	Original correct	Reminder correct	Gains / losses	Difference (pp)	Conditional 95% paired bootstrap interval (pp)
Qwen3-4B	24/49	20/49	3 / 7	-8.16	[-20.41, 4.08]
Qwen2.5-7B	14/49	15/49	4 / 3	+2.04	[-8.16, 12.24]
Qwen3-30B-A3B	27/49	29/49	4 / 2	+4.08	[-4.08, 14.29]

For this descriptive analysis, we compute the exact finite distribution of the stratified paired bootstrap. Each task pair contributes -1, 0 or +1 for reminder-minus-original correctness. Within each of the four fixed domains, we resample its observed number of scored pairs with replacement, then sum over domains and divide by 49. Convolution with rational probabilities gives the 2.5th and 97.5th percentiles without Monte Carlo approximation. A small exhaustive-resampling check validates this calculation. This is an ordinary conditional bootstrap calculation, not a new estimator or a preregistered hypothesis test. It does not cover unseen-domain sampling, model randomness or annotation uncertainty, and we make no multiplicity-adjusted superiority claim across checkpoints.

Every interval contains zero. Omitting Disney gives a net difference of -1/31 tasks (-3.23 points) for each checkpoint, reversing the two positive full-grid differences. Applying the eight previously disclosed conservative Disney label changes, without adding new changes, produces counts 21/49 versus 19/49, 14/49 versus 14/49, and 25/49 versus 28/49 respectively; all three corresponding conditional intervals still include zero. These checks restrict the aggregate prompt-benefit interpretation. They do not establish prompt equivalence or erase the within-domain gains and losses.

4.2 Sensitivity to answer annotation

A separate post-hoc calculation addresses label sensitivity rather than task resampling. We group exact final-answer text within the same question, across prompts and checkpoints, and require one shared requested-value judgment for every such group. The 364 present responses form 348 distinct question/text groups; fifteen repeated groups cover 31 executions, with no conflicting labels. This consistency check does not adjudicate correctness. Missing final answers remain fixed failures. Flipping a group's binary judgment changes all its occurrences together. Under this deliberately hypothetical model, Qwen2.5's one-answer gain can be reduced to a tie by one changed judgment and reversed by two; Qwen30B's two-answer gain requires two and three changes, respectively. Qwen3's four-answer loss requires four changes to reach a tie and five to reverse. These are minimum hypothetical changes, not discovered annotation mistakes, error probabilities or a second reviewer's decisions. Per-checkpoint extremal assignments need not be simultaneously attainable across checkpoints.

The 21 ambiguous questions create a different boundary. For a hypothetical 70-question score, hold all 49 primary-task labels fixed and allow arbitrary binary labels only for present ambiguous answers, tying exact duplicate text and keeping absent answers unsuccessful. The resulting reminder-minus-original net-difference envelopes are [-21, 13], [-15, 17] and [-10, 18] for Qwen3, Qwen2.5 and Qwen30B. These are conservative label-relaxation envelopes, not population intervals or sharp bounds over coherent SQL interpretations; some extremal label assignments may have no common semantic interpretation. They leave the original denominator and labels unchanged and reinforce that the reported conditional 49-task comparison is not a score for all 70 tasks.

4.3 Resource limits and development context

Forty-three expansion episodes terminate at the input ceiling, all in ice hockey. Saved attempted-input counters range from 32,849 to 161,160 tokens, after one to four successful generations. These are trace counters, not independently retokenized measurements. No failure is discarded from a scored denominator or from execution costs.

Disney illustrates the judgment boundary: some responses give a correct count with wrong supporting film names, or name the required values while adding contradictory statements. The eight-case sensitivity above changes those specific judgments; it is not a bound on all possible annotation error.

Earlier known-task comparisons provide context rather than independent confirmation. Strict Qwen3 changes from 29 to 32 correct answers out of 55; strict Qwen2.5 changes from 19 to 21. Qwen3's difference reverses sign when cars is removed. Qwen3-30B changes from 35 to 38, with a conditional interval of [-3.64, 14.55] percentage points. SmolLM3 yields only six correct answers among 220 scored executions, despite 221 final responses among 240 runs. Its floor and 66 finals at the output ceiling limit cross-family inference.

5. RQ2: Executor restrictions and output capacity

5.1 Relaxing a call restriction

The strict executor contributes 99 of 128 protocol-error events in the three-domain comparison. Repeating its full 240-episode grid with sequential batch execution holds tasks, models, prompts and primary budgets fixed. All first replies match; 37 trajectories diverge after model inputs change, while 213 final answers remain byte-identical. This is a known-task sensitivity study, not a replacement for the original results.

Table 4. Final-response production and correctness under two executors.

Checkpoint / prompt	Strict finals	Sequential finals	Strict correct	Sequential correct
Qwen3 / original	60/60	60/60	29/55	29/55
Qwen3 / reminder	60/60	60/60	32/55	32/55
Qwen2.5 / original	51/60	56/60	19/55	19/55
Qwen2.5 / reminder	49/60	56/60	21/55	22/55

Twelve additional Qwen2.5 final responses produce one net additional correct answer. Recovered answers can omit rows, misalign fields or use the wrong scope. Under the original prompt, one correct recovered answer is offset by loss of another; the reminder contributes the net gain.

Across Qwen2.5 prompts, protocol errors decrease from 127 to 31 and model attempts from 542 to 498, while tool attempts increase from 315 to 455. Input tokens decrease from 6,158,394 to 5,693,901 and output tokens from 54,925 to 53,818. These opposing resource changes prevent an unqualified efficiency claim. A concurrent download prevents an isolated wall-time comparison.

5.2 Increasing the generation ceiling

Before predictions, reference inspection selected complete lists of 791 alcohol-free recipes and 129 hockey players. Three explicit serializations with the pinned Qwen30B tokenizer measure 5,047-5,406 and 646-649 tokens, respectively. These motivate a registered control but are not lower bounds on every valid answer, nor measurements of the other tokenizers.

The twelve prospectively registered controls complete, with every worker exiting successfully. The only configured budget change is 512 to 8,192 new tokens on every target generation. Source versions, packages, model files, templates, decoding, input limits and tool/step ceilings match the corresponding primary configurations. Before target generation, the runner replays each worker's warmup and preceding tool history without model inference and checks the initial messages, ordered tool schemas and preview against the primary record. Offline comparison independently confirms those initial inputs. All twelve runs produce final answers, without protocol, input-limit or step-limit terminations; none reaches the new generation ceiling.

Table 5. All twelve registered generation-budget controls.

Task / checkpoint	Primary complete answers, original / reminder	Larger-budget complete answers, original / reminder	Final tokens, larger-budget original / reminder
791 recipe names / Qwen3-4B	0 / 0	0 / 0	55 / 94
791 recipe names / Qwen2.5-7B	0 / 0	0 / 0	60 / 86
791 recipe names / Qwen3-30B-A3B	0 / 0	0 / 0	81 / 85
129 player names / Qwen3-4B	0 / 0	1 / 1	617 / 619
129 player names / Qwen2.5-7B	0 / 0	0 / 0	33 / 54
129 player names / Qwen3-30B-A3B	0 / 0	1 / 1	626 / 628

All four recovered player lists contain exactly the 129 requested names, checked by parsing their complete comma-separated bodies and comparing name multisets with the frozen SQL card. The corresponding short-budget answers end at 512 generated tokens and omit required names. The eight other final responses are byte-identical to their primary answers: they still provide only the first three entries, sometimes offering a future retrieval or mentioning the total. This is insufficient under the unchanged complete-list criterion. All twelve full responses are also read by the same unblinded assistant; the mechanical check is not an independent semantic adjudication.

The observed match extends beyond initialization. All twelve pairs have identical pre-final model inputs and generated token-ID sequences, identical ordered tool names/arguments and response content/error flags, and identical inputs to the final generation. Each short final's token IDs are an exact prefix of the corresponding long final's IDs. Thus, in these four recovered cases, the larger allowance extends the same observed final generation after the same acquisition history. This supports a local attribution to the output cap, stronger than merely noticing a response at the ceiling. The intervention nevertheless changes all target generations; equivalent pre-final behavior is an observed result here, not a guarantee for other tasks or checkpoints.

This contrast separates two limited explanations. Four player-list failures are relieved by a larger generation allowance, whereas the eight unchanged responses stop far below that allowance with incomplete lists. It does not

identify why those agents stop, establish that all needed evidence was acquired, or demonstrate an autonomous acquisition repair. Two selected questions with six model/prompt arms are not twelve independent task samples. No confidence interval for a general recovery rate is inferred, and the twelve controls do not replace or rescore the primary 420-run experiment.

5.3 Available evidence is not necessarily acquired

Cookbook task 015 provides a complementary control. Its six histories reach the correct maximal-cooking-time filter but fail to list all ten ingredients. An evaluator-authored native-tool continuation sorts names and repeatedly filters above the last visible name. Ordinary keyset pagination exposes remaining row counts 10, 7, 4 and 1 and recovers the complete answer in four added calls. Including unsuccessful prefix attempts, total target calls are six or seven, below the twenty-call ceiling.

Warmup, 224 preceding responses and the six target prefixes replay exactly; the database hash remains unchanged. The continuation uses only tool replies; offline SQL validates its result afterward. It requires stable data, non-null strings, compatible ordering and accurate row counts. This proves feasible acquisition for one selected question, not autonomous repair or a general solution: the evaluator chose the checkpoint and column after review, and no model continued from the repaired observations.

6. RQ3: Correct answers under indistinguishable observations

6.1 Two retained witnesses

In a world-database episode, Qwen3 with the reminder correctly returns Catalan for the official language of Andorra, the highest-life-expectancy country. Its four-row handle previews only three language rows. Changing the unseen Spanish row's official-status flag preserves initialization, ordered schemas and both complete recorded tool responses while changing the SQL answer to Catalan and Spanish. All other data stay fixed. An earlier independent episode would expose the changed value but is absent from the target's fresh model messages. This claim is episode-local and conditional on allowing that status change, not on immutable real-world language knowledge or cross-task memory.

The publishing witness removes an apparent remedy: full output retrieval. Qwen3's original-prompt episode selects authors with contract values unequal to γ , then retrieves both complete title and sales columns. Its seventeen distinct pairs match the released data, where all 23 author flags are zero. Under the explicit intervention admitting one flag change from 0 to 1, the required answer loses one title, but all target observations remain identical: both zero and one differ from γ . Complete output retrieval does not establish the predicate's correctness.

An evaluator-supplied equality filter with value 0, followed by the same getters, returns seventeen versus sixteen pairs and agrees with SQL in both worlds. The interface can distinguish the states; the recorded agent did not. This CPU control is not a new neural run or evidence of autonomous repair. The positive witness is one of six from eight contract-related episodes; the other five do not begin with correct answers.

6.2 Negative and restricted controls

A broad world-database search retains at most eight answer-changing candidates from 2,048 seeded draws per task. Forty original trajectories replay exactly; 58 mutant databases and 141 replays yield seventeen observation-preserving witnesses. Four involve correct original answers, but change geographical membership, such as assigning a Japanese district to England. We reject them as support for the retained claim: schema validity does not preserve the background constraints used to interpret those questions.

An exhaustive narrower audit of 984 official-language flags recovers the known Spanish-status witness. Of six witnesses across eight target episodes, only that known case begins with a correct answer. This is a positive replay control, not independent evidence of prevalence.

In cars, swapping two existing positive prices preserves their multiset, other cells and declared keys. Across twenty tasks and eighty sequential executions, 99 answer-changing mutants and 332 replays yield ten witnesses, all for already wrong answers. None of the 43 correct-answer episodes gains a witness; seven tasks have no retained candidate. This failed extension limits generalization, while bounded search failure certifies no answer.

Two `computer_student` mutations preserve prescribed initial relations while changing answers under the documented advisor relationship. Indistinguishability holds only for operations on those relations and descendants. The live API accepts other universe identifiers, and 14 of 27 alternative initializations expose different full relations under each mutation. We make no impossibility claim about the unrestricted interface or agents with external knowledge.

7. Related work

[Agents Don't Paginate](#) studies first-chunk selection and a file-localization probe, including negative downstream results. [Fabrication After Tool Failure](#) studies reporting after unusable payloads and prompt-level defenses. [How Good Are LLMs at Processing Tool Outputs?](#) compares answer generation, executable processing and response representations. Our reminder belongs to this established intervention space. Our controls instead examine continued relational acquisition, executor constraints and answer capacity together; they do not demonstrate superiority over executable processing methods.

[Blockaid](#) formalizes trace-conditioned database determinacy. [Distilled test suites](#) distinguish SQL programs that agree accidentally on one database. Our replays apply related reasoning to complete interactive observations, requiring admitted interventions to leave the agent-visible history unchanged. [GroundEval](#) uses state-dependent evidence contracts; [HERALD](#) studies counterfactual retrieval audits; [EG-VAR](#) derives proof-backed claims through a trusted data-to-logic boundary. The present empirical controls and local witnesses do not introduce formal grounding or counterfactual auditing.

[ELT-Bench-Verified](#) investigates benchmark defects with automated diagnosis and human validation, while [AgentJudgeBench](#) evaluates judges of tool calling. Frozen interpretation cards and disclosed masks are safeguards here, not independent adjudication or evidence that our labels outperform an official evaluator. Full bibliographic records accompany the manuscript.

8. Discussion and limitations

These controls establish different boundaries. Relaxing call execution mainly recovers interaction; increasing generation capacity recovers four exact continuations but leaves early-stopped previews unchanged. Full output retrieval can still miss a predicate error. Evaluation should report these levels separately, with resources and counterfactual assumptions explicit. No single dominant failure cause follows across all tool-using agents.

Tasks come from one benchmark's public training material and eight databases selected partly for local feasibility. The new-domain comparison is prospective relative to our experiments, but development informed model and intervention choices. Four checkpoints do not represent a population of models. Each cell has one greedy run; intervals describe task composition rather than repeated generation. Oracle initialization, local templates and budgets constrain comparisons with hosted agents and official scores.

Answer labels are assistant-authored and unblinded. Fixed cards reduce outcome-dependent reinterpretation but cannot remove reviewer bias. Some credited answers have faulty explanations. After the earlier student-domain review, an additional ambiguity between employee ID and job level was noticed; the frozen job-level interpretation was retained and disclosed. The Disney sensitivity examines eight specific judgments, not all possible label errors. The additional label-flip and ambiguity envelopes expose sensitivity under explicit hypothetical assignments; they neither estimate annotation error rates nor replace independent adjudication. Hash agreement verifies provenance, not semantic truth.

Replay examples are post-hoc and conditional on synthetic status changes. They establish existence, not incidence, and do not exclude helpful priors or further distinguishing interactions. Pagination and predicate repairs are evaluator-selected controls, not an end-to-end agent method. Hardware and download overlap prevent a common speed ranking. More recorded executions alone do not establish a new controller or benchmark-wide prevalence.

9. Conclusion

Interaction completion, answer agreement and identification by observed evidence behave differently in these relational-tool studies. The fixed reminder has mixed cross-domain effects. A permissive executor mainly increases final-response production; larger generation budgets resolve four demonstrably truncated lists but leave eight incomplete responses unchanged. Two native-tool witnesses show how correct answers can remain unidentified within explicit alternative worlds, including after full output retrieval. Retained inadmissible and unsuccessful searches keep these claims local. Evaluation of tool use needs joint attention to scope, acquisition, answer capacity and the evidential meaning of the trace.

Data, code and use of language models

Sources are maintained on the `codex/agent-paper-research` branch of [Jing-XING/autoresearch](#). `reproduce.md` documents the complete expansion and budget analyses; `claim_to_evidence.md` links earlier claims to their artifacts. Raw archives are retained locally with hashes and are not yet a complete public download. `study_record.md` preserves the preceding manuscript verbatim, including protocol history and detailed controls. `references.bib` supplies the cited bibliographic records.

A language-model assistant performed literature retrieval, implementation, experiment operation, answer review, analysis and drafting. Numerical results come from logged executions and retained artifacts. Answer annotation has no independent human review. Author identities, affiliations, funding, accountability declarations and submission approval have not been invented or assigned by the assistant.

Appendix A. Exact reminder

Before answering, verify that the observed results cover the requested scope. A `first_3_values` preview is a sample, not a complete column. Compare it with `num_records`. For a complete list, obtain the relevant full column from a correctly filtered handle. For a count or aggregate, use the appropriate tool on the correctly filtered data. Filter or aggregate large tables before fetching full columns. Check every requested condition, including any status qualifier. If the evidence remains incomplete, explicitly state that limitation instead of claiming completeness.

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